Truncations of the Higman–Sims geometries

L091 classification component

We use the complete enumeration and the separate proof of bare incidence isomorphism classes for the specified Higman–Sims action. There are $1, 6, 30, 41, 8, 1$ classes in ranks one through six and no higher ranks. A truncation below means deletion of types from another geometry in this classified family. The internal representative numbers refer to the fixed file `full_fusion03/result.g`.

Proposition 1. *Exactly two rank two classes and ten rank three classes are truncations of higher rank geometries in the family. No class of ranks four, five or six is such a truncation. The sole rank one class is a truncation. Consequently the numbers of classes which are not truncations are*

$$0, \quad 4, \quad 20, \quad 41, \quad 8, \quad 1.$$

Proof. Let G have order $44\,352\,000$, and let $(G_i)_{i\in I}$ be a base chamber's parabolic stabilizers. For a subset $J \subseteq I$, the truncation is the coset geometry with parabolics $(G_i)_{i\in J}$. It inherits flag transitivity. Thus its type sizes, pair-flag counts and chamber count determine respectively the multisets

$$\{|G_i| : i \in J\}, \qquad \{|G_i \cap G_j| : \{i, j\} \subseteq J\}, \qquad \left|\bigcap_{i\in J} G_i\right|.$$

These are necessary invariants for incidence isomorphism with any member of the classified list, since the group order is fixed. Their use does not assume that an incidence isomorphism normalizes G.

For every one of the 86 representatives of rank at least two, take every proper subset J with $|J| \geq 2$. There are exactly

$$30\binom{3}{2} + 41\left(\binom{4}{2} + \binom{4}{3}\right) + 8\left(\binom{5}{2} + \binom{5}{3} + \binom{5}{4}\right) + \binom{6}{2} + \binom{6}{3} + \binom{6}{4} + \binom{6}{5} = 756$$

such subsets. For each, compare the displayed invariant with every representative. Exactly fifteen comparisons survive. Each has a saved permutation in the acquired full automorphism action of G and a permutation of types, sending the actual truncation parabolics onto those of its target. These give explicit coset isomorphisms. There are no surviving comparisons without a positive witness. The distinct targets are $3, 4$ in rank two and $8, 13, 18, 20, 21, 22, 28, 29, 30, 31$ in rank three.

An arbitrary classified parent is incidence-isomorphic to a listed representative; any such isomorphism transports its type subsets. Hence the finite list of comparisons covers every possible parent and truncation. Every excluded comparison fails an intrinsic incidence invariant, while every surviving one is positively identified. This proves completeness of the claimed twelve targets, including the negative assertions in ranks four through six. The separately checked rank one component supplies a truncation witness for its unique 176-object geometry. Subtraction from the complete rank counts proves the displayed result. □

Attribution and computational scope. These are the truncation totals stated in D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Completeness of the parent classification is an explicit acquired premise.

The new finite comparison uses actual subgroup intersections and the acquired simultaneous conjugation method in GAP; every positive witness is checked directly. No inference uses failure of a conjugator search. This component supplies the truncation clause, not the supplementary diagrams, automorphism groups or the two geometric constructions, and claims no new open result.